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Inventor(s)

Redzematovic; Mersim et al.

ROBOTIC SPALL DAMAGE REPAIR SYSTEM AND METHOD

Abstract

A system and method for repairing one or more spalls using an autonomous robotic platform is provided. The system for repairing one or more spalls can include a power module, a construction module, a mechanical module, and a processing module. The method for repairing one or more spalls can include: processing coordinates and navigating to a spall, preparing the spall for repair, measuring a volume of the spall and determining an amount of repair material to be used, dispersing the appropriate amount of repair material into the spall, finalizing the repair of the spall, and generating a notification.

Inventors: Redzematovic; Mersim (Lynbrook, NY), Senft; John (Babylon, NY), Caradonna; Austin (Massapequa, NY), Asmari; Ali (Selden, NY)

Applicant: ULC Technologies LLC (Hauppauge, NY)

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Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/563,778 filed Mar. 11, 2024, the entire disclosure of which is incorporated herein by reference.

FIELD OF THE INVENTION

[0002] This disclosure generally relates to autonomous robots. More specifically, this disclosure relates to a system and a method for repairing spalls using an autonomous robot.

BACKGROUND OF THE INVENTION

[0003] A spall includes a surface that is broken, cracked, or otherwise damaged. Spalling damage can occur in roadways, bridges, parking areas, airfields, buildings, or any other structure or surface created from concrete, brick, stone, asphalt, metal, or similar construction material. Conventional systems and methods for repairing spalls utilize heavy and complex equipment, often multiple pieces of equipment, along with multiple operators who are highly-skilled and crossed-trained. Furthermore, deploying and maintaining such large, specialized equipment, can be challenging and costly, especially in remote areas. In some instances, a spall may be located in a military zone, on an active airfield, or other area that may pose a safety risk to human operators repairing the spall. Therefore, there is a need for an automated process to repair spalls without the use of manual labor and heavy machinery.

BRIEF SUMMARY OF THE INVENTION

[0004] An aspect of this disclosure pertains to a method and system for repairing one or more spalls using an autonomous robotic platform.

SUMMARY

[0005] An autonomous robotic platform for repairing a spall is provided. The autonomous robotic platform comprising: a power module; a construction module, including a tank for holding a repair material, a mechanical module, including a track component for moving the autonomous robotic platform; and a plow attachment, for moving debris proximate to the spall to be repaired. The autonomous robotic platform continuing to include a processing module comprising: a vision system module, designed to identify and analyze the spall; a control module, to execute one or more programmable instructions of the autonomous robotic platform; and a navigation module.

[0006] In some aspects, the processing module further comprises: a notification module, a sensor module, a communication module, and an electrical safety system. In some forms, the vision system module comprises: a camera system, a sensor system, and one or more computer vision algorithms. In another aspect, the navigation module is configured to process location coordinates associated with a spall repair assignment and navigate the autonomous robotic platform to the location coordinates. In some aspects, the control module further comprises a disconnect switch. In some forms, the power module includes a generator, a battery, a battery management system, or a combination thereof. In another aspect, the notification module sends a release signal to indicate that the autonomous robotic platform has reached the location coordinates. In some aspects, the electrical safety system is configured to limit operations of the autonomous robotic platform in a defined three-dimensional work zone for a spall repair assignment.

[0007] An autonomous robotic platform designed for repairing one or more spalls is provided. The platform comprising: a power module; a construction module, including a tank for holding a repair material and a nozzle for dispensing the repair material; a mechanical module, including a track component, for moving the autonomous robotic platform; and a processing module. The processing module comprising: a vision system module, designed to identify and analyze the spall; and a navigation module, designed to route the autonomous robotic platform to a location coordinate associated with the spall.

[0008] In some aspects, the power module includes a generator, a rechargeable battery, a battery, a battery management system, or a combination thereof. In some forms, the mechanical module comprises: a plow attachment, for moving debris proximate to the spall to be repaired; and a plate compactor attachment to tamper the spall after the repair material is applied to the spall. In some examples, the tank for holding repair material is provided in the form of a liquid tank, a binding agent tank, a hopper, or a combination thereof. In some aspects, the autonomous robotic platform further comprises: a user interface, to send and retrieve information from a data store; and a network, communicatively coupled to the autonomous robotic platform.

[0009] A method of repairing a spall using an autonomous robotic platform is provided. The method providing location coordinates associated with a spall repair assignment to a navigation module of the autonomous robotic platform. The method navigating to the location coordinates using a vision system module. The method preparing the spall for repair using a plow attachment to move debris proximate to the spall. The method analyzing a volume of the spall using the vision system module. The method determining an amount of repair material to be used for the spall. The method dispersing the amount of repair material determined into the spall and finalizing the repair assignment. The method then generates a notification to a user device that the spall repair assignment is complete.

[0010] In some aspects the location coordinates are provided in the form of GPS coordinates. In some forms, the method applies a liquid to the spall from a liquid tank of the autonomous robotic platform. In some examples, the method prepares the spall for the repair further comprising moving the debris into the spall and analyzing the volume of the spall after the debris has been moved into the spall. In some aspects, the method prepares the spall for repair includes flooding the spall with liquid. In some forms, the liquid is water. In some examples, the notification includes metadata and an image of the spall after the repair.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings, which are incorporated in and form a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of embodiments of the invention:

[0012] FIG. 1 is an isometric view of a front of an autonomous robotic platform;

[0013] FIG. 2 is a plan view of a top of the autonomous robotic platform of FIG. 1;

[0014] FIG. 3 is an elevational view of a rear of the autonomous robotic platform of FIG. 1;

[0015] FIG. 4 is an elevational view of a side of the autonomous robotic platform of FIG. 1;

[0016] FIG. 5 is an elevational view of a front of the autonomous robotic platform of FIG. 1;

[0017] FIG. 6 is a plan view of a bottom of the autonomous robotic platform of FIG. 1;

[0018] FIG. 7 is a block diagram of modules included in the autonomous robotic platform of FIG. 1;

[0019] FIGS. 8A-8F illustrate a method for spall repair using the autonomous robotic platform of FIG. 1; and

[0020] FIG. 9 is a flowchart of a process for repairing spalls using the autonomous robotic platform of FIG. 1.

[0021] Before explaining the disclosed embodiment of the present invention in detail, it is to be understood that the invention is not limited in its application to the details of the particular arrangement shown, since the invention is capable of other embodiments. Exemplary embodiments are illustrated in referenced figures of the drawings. It is intended that the embodiments and figures disclosed herein are to be considered illustrative rather than limiting. Also, the terminology used herein is for the purpose of description and not for limitation.

DETAILED DESCRIPTION

[0022] The following discussion is presented to enable a person skilled in the art to make and use embodiments of the invention. Various modifications to the illustrated embodiments will be readily apparent to those skilled in the art, and the generic principles herein can be applied to other embodiments and applications without departing from embodiments of the invention. Thus, embodiments of the invention are not intended to be limited to embodiments shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein. The following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. The figures, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of embodiments of the invention. Skilled artisans will recognize the examples provided herein have many useful alternatives and fall within the scope of embodiments of the invention.

[0023] Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the attached drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. For example, the use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

[0024] As used herein, unless otherwise specified or limited, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, unless otherwise specified or limited, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

[0025] As used herein, unless otherwise specified or limited, “at least one of A, B, and C,” and similar other phrases, are meant to indicate A, or B, or C, or any combination of A, B, and/or C. As such, this phrase, and similar other phrases can include single or multiple instances of A, B, and/or C, and, in the case that any of A, B, and/or C indicates a category of elements, single or multiple instances of any of the elements of the categories A, B, and/or C.

[0026] Some embodiments include an automated system including one or more robotic platforms for repairing spalls. Some embodiments include advanced navigation processes to generate and execute a spall repair strategy. In some embodiments, the robotic platform may work in tandem one or more other systems, subsystems and/or operators to complete a repair of a spall in specific locations. In some embodiments, the automated robotic platform described herein may repair one or more spalls and travel to a location of the next spall to repair.

[0027] In some embodiments, a sequence of one or more actions performed by the robotic platform may be controlled by a release signal (or similar) via a communication module, to indicate that the autonomous robotic platform robot is no longer moving or is otherwise aligned in accordance with the spall repairing process, and that the next phase of repair may begin. In some embodiments, a sensor module can communicate with a control module to perform the steps of repairing one or more spalls. In some aspects, the sensor module can initiate the next phase of repair based on determining that the previous repair step has been completed, or does not need completed, as determined using one or more aspects of a processing module of the automatic robotic platform system. In some embodiments, the process of determining a location for repair of spall and completing the repair spall process can be iteratively repeated until the spall is repaired.

[0028] In some embodiments, the robotic platform can further include one or more modules or subassemblies including, but not limited to: a notification module, a control module, a safety system module, a sensor module, an electrical module, a command module, a navigation module, a communication module, a vision system module, a power module, a construction module, a

mechanical module, and other modules and subsystems. In some forms, the power module includes a generator, rechargeable battery, a battery management system, or a combination thereof. In some embodiments, other modules or subassemblies are included.

[0029] FIG. 1 illustrates a non-limiting example of an autonomous robotic platform **100**. The autonomous robotic platform **100**, a hopper **105** for containing a local aggregate **106**, and one or more hopper supports **107**. The autonomous robotic platform **100** further comprises a liquid tank **108**, a liquid tank lid **109**, a tank drain **110**, a binding agent tank lid **111**, a binding agent tank **112**, a track component **113**, a plow attachment **114**, a binding agent nozzle **115**, a liquid nozzle **116**, an auger **117**, an auger tube **118**, a main base plate **119**, and one or more tubes **120**.

[0030] In some embodiments, the local aggregate **106** can be dirt, soil, mud, clay, sand, crushed stone or concrete, slag, recycled materials, gravel, or combination thereof. In some aspects, the hopper **105** can be configured to hold other bulk materials, other than local aggregate **106**, if different materials are utilized for the repair according to the spall repair assignment and/or the repair material specifications. For example, some repair materials and/or binding agents may not specify to use a local aggregate with the material, in which case the hopper **105** can be disabled for the particular repair assignment, or the hopper **105** can be utilized to hold a different type of bulk material). In some embodiments, the hopper **105** can be configured to include a lid or covering to contain or otherwise protect the bulk materials from the environment (e.g., water, wind, the sun, etc.). In some embodiments, the hopper **105** is constructed of a durable material suitable for the type(s) of liquid or material being stored in the hopper **105**. As shown in FIG. 6, the hopper **105** can be coupled to a hopper mount plate **605**. The hopper mount plate **605** can be mounted to a frame of the autonomous robotic platform **100** and used to secure the hopper **105**.

[0031] The auger **117** and the auger tube **118** may be coupled to the hopper **105** to convey the local aggregate **106** to the spall being repaired. In some forms, the local aggregate **106** is dispensed to the spall via the auger tube, which can be provided in the form of a tube, chute, or similar device for transferring the local aggregate **106** from the hopper **105** to the spall. In some aspects, the auger **117** can be provided in the form of a spiral drill, or similar device, used to excavate portions of the spall or surrounding areas to prepare the spall for repair.

[0032] The liquid tank **108**, in some examples, can be configured to contain water or other liquids or materials. In some aspects, the liquid tank **108** can be configured to hold other materials, other than water, if different materials are utilized for the repair according to the spall repair assignment and/or the repair material specifications. For example, some repair materials and/or binding agents may not specify to use water or other liquid with the repair material, in which case the liquid tank **108** can be disabled for the particular repair assignment, or the liquid tank can be utilized to hold a different type of material). In some embodiments, the liquid tank **108** is constructed of a durable material suitable for the type(s) of liquid or material being stored in the liquid tank **108**.

[0033] The binding agent tank **112**, in some examples, can be configured to contain a binding agent, glue, resin, sealant, asphalt, polymer, cement, a binder, or other types of repair materials. In some aspects, the binding agent tank **112** can be configured to hold other materials, other than water, if different materials are utilized for the repair according to the spall repair assignment and/or the repair material specifications. For example, some repair materials and/or spall repair specifications may not specify to use a binding agent to complete the repair, in which case the binding agent tank **112** can be disabled for the particular repair assignment, or the binding agent tank **112** can be utilized to hold a different type of material). In some embodiments, the binding agent tank **112** is constructed of a durable material suitable for the type(s) of liquid or material being stored in the liquid tank **108**.

[0034] The track component **113** is configured to propel and maneuver the autonomous robotic platform **100** and may be positioned and located on opposing sides of the autonomous robotic platform **100**. The track component **113** may be provided in the form of wheels, rollers, legs, gliders, treads, propellers, thrusters, or other forms of movable support for traversing the robotic

platform **100**. As will be described in further detail in connection with FIG. 7, the track component **113** may be controlled by a control module **706** to drive, steer, or otherwise maneuver the autonomous robotic platform **100**.

[0035] The plow attachment **114**, in some embodiments can be configured to be switched out for different attachments for manipulating a surface area in/around a spall. In some aspects, the plow attachment **114** can be provided in the form of a bucket, spoon, scraper, air hose end effector, vacuum, or other device used to move debris, clean a surface, or prepare a spall for repair.

[0036] In some examples, the binding agent nozzle **115** and the liquid nozzle **116** may be provided in the form of a convergent nozzle, a divergent nozzle, a straight jet nozzle, a fan nozzle, a hollow cone nozzle, a full cone nozzle, an air atomizing nozzle, a swirl nozzle, a pressure-swirl nozzle, a venturi nozzle, a fog nozzle, a multi-orifice nozzle or combination thereof.

[0037] As shown in FIG. 5, the binding agent nozzle **115** and the liquid nozzle **116** can be secured by a nozzle bracket **505**. The nozzle bracket can be coupled to, fastened around, or otherwise securely fixed to the one or more hoses **120** connected to the binding agent nozzle **115** and the liquid nozzle **116** to aim the nozzles in a particular direction relative to the repair location. In some aspects, the platform **100** may be configured to control an orientation and/or position of the binding agent nozzle **115** and/or the liquid nozzle **116** based on one or more parameters associated with the repair specification. For example, if the repair specification for a particular repair uses a bonding agent that should be applied with minimal spray and/or residue, then the platform **100** may use the control module **706** (or other module) to adjust the position of the binding agent nozzle **115** to be located closer to the ground and at a more direct angle to the repair area, to minimize overspray.

[0038] The autonomous robotic platform **100** can further be configured for different use cases. For example, the autonomous robotic platform **100** can have not only three material-carrying modules (e.g., the hopper **105**, the liquid tank **108**, and the binding agent tank **112**) but also more or less modules of different sizes for the autonomous robotic platform **100**. The autonomous robotic platform **100** further can communicate with one or more autonomous robotic platforms and work collaboratively via wireless communication protocols, using a communication module **707** further disclosed in FIG. 7. The autonomous robotic platform **100** can communicate with one or more external servers and one or more operators remotely in real time via the communication module **707** further disclosed in FIG. 7.

[0039] The autonomous robotic platform **100** may also include a user interface for an operator **104** to interact with the system **100**. In some aspects, the user interface can be provided with a sunshade or another sun-blocking or glare-reducing devices. In some forms, the user interface can include photodiodes to provide automatic dimming in sunny conditions in order to make the screen more readable. In some aspects, the user interface can include interactive elements for the operator **104** to select or otherwise manipulate (e.g., buttons, switches, icons, drop down menus, links, etc.). Examples of features and operations accessible via the user interface may include settings, alerts, alarms, notifications, maps, status updates, process diagrams, queues, user input fields, reports, lights, haptics, etc. The user interface, in some aspects, may generate and transmit notifications to the operator **104** and/or the one or more external servers or user devices for routine maintenance, recommended repairs, and preventative maintenance.

[0040] The autonomous robotic platform **100** may include one or more lights which are located around the perimeter of the platform to provide an operator **104** with information including visual alerts. In some embodiments, the autonomous robotic platform **100** can include one or more work lights mounted around the perimeter of the platform to improve visibility of the device and/or an operator, particularly for nighttime spall repairs.

[0041] FIGS. 2-6 illustrate alternative views of the autonomous robotic platform **100** of FIG. 1. For instance, FIG. 2 illustrates a top view of the autonomous robotic platform **100** including the operator **104**, local aggregate **106**, the liquid tank **108**, the tank lid **109**, the binding agent tank lid **110**, the binding agent tank **112**, the track component **113**, the binding agent nozzle **115**, the liquid

nozzle **116**, and the auger **117**. FIG. **3** illustrates a back view, FIG. **4** illustrates a side view, FIG. **5** illustrates a front view, and FIG. **6** illustrates a bottom view of the autonomous robotic platform **100** of FIG. **1**.

[0042] Additionally, FIG. **2** illustrates examples of the binding agent spray area **121**, and the liquid spray area **122**. In some embodiments, the example spray areas can be expanded or narrowed or otherwise adjusted based on the repair specifications and/or the materials being ejected from the nozzles. In some embodiments, the spray areas can be automatically adjusted using the control module **706** of the system, or remotely by the operator **104** or other user device.

[0043] The autonomous robotic platform **100** further includes a generator **205**. The generator **205** can be configured to be powered by fuel, electricity, or combination thereof. In some aspects, the autonomous robotic platform **100** may include a rechargeable battery. In some aspects, the generator **205** may be configured to work together with a rechargeable battery integrated with the autonomous robotic platform **100**. In some examples, the generator **205** and/or battery, may be configured to be removed easily for maintenance, repair, refueling, recharging, etc. The user interface may be configured to display the overall health, power levels, and system runtime of the rechargeable battery and/or the generator **205** using a sensor module **712**, further described in connection with FIG. **7**.

[0044] As shown in FIG. **3**, the autonomous robotic platform **100** further includes a processing module **305**. The processing module **305** includes one or more modules and/or submodules, as discussed in connection with FIG. **7**. The processing module **305** is configured to communicate wired or wirelessly to the one or more modules via an electrical module **714** (see FIG. **7**).

[0045] The autonomous robotic platform **100** further includes a motor **306** and a plate compactor attachment **307** (e.g., a tamper). The motor **306** can be configured to be operated using fuel, electricity, hydraulics, or combination thereof. The plate compactor attachment **307** can be configured to include one or more sensors to determine when a ground surface associated with the repair is level, before determining a repair is complete. In some examples, the plate compactor attachment **307** can be easily removed for maintenance or repair. In some aspects, the plate compactor attachment **307** can be modular and be swapped with other attachments or accessories, depending on the type of repair to be completed, the type of surface being repaired, or other parameters. The plate compactor attachment **307** can be configured to move vertically or horizontally, which is particularly useful when leveling a surface area during a repair process, as discussed in more detail in connection with FIG. **8F**. Additionally, the autonomous robotic platform **100** includes one or more track structural beams **308** and one or more frame base beams **309**. The one or more track structural beams **308** mechanically couple the track components **113** to a frame of the autonomous robotic platform **100**. The one or more frame base beams **309** support the main base plate **119** of the autonomous robotic platform **100**.

[0046] FIG. **7** illustrates a block diagram of system components **700** of the autonomous robotic platform **100** of FIGS. **1-6**. The system components **700** include a processing module **705**, a power module **708**, a construction module **709**, and a mechanical module **710**.

[0047] The processing module **705** includes the control module **706**, the communication module **707**, a notification module **711**, a sensor module **712**, a navigation module **713**, an electrical module **714**, a safety system module **715**, a command module **716**, and a vision system module **717**.

[0048] The control module **706** is designed to execute commands or programmable instruction to perform one or more tasks assigned the autonomous robotic platform **100**. For example, the control module **706** is designed to execute commands or programmable instructions to facilitate repairs of one or more spalls on a street, roadway, a military zone, a parking lot, an airfield strip, or other surface. In some embodiments, the control module **706** is designed to operate fully autonomously to perform or otherwise execute one or more tasks (e.g., repair identified spalls on a roadway or on an airfield strip). In other embodiments, the operator **104** may supervise the one or more repairs of

the one or more spalls and/or remotely control the repair operations. In some embodiments, the control module **706** includes one or more controllers for the autonomous robotic platform **100**. In some aspects, the autonomous robotic system **100** can be controlled using a user device, an unmanned aerial system (UAS), a cloud server, or other computing device.

[0049] The control module **706** is generally designed to control the operation of one or more aspects of the autonomous robotic platform **100**. (See FIGS. 1-6). More particularly, the control module **706** may include a processor designed to execute programmable instructions to operate one or more aspects of the autonomous robotic platform **100** in order to perform various actions associated with repairing spalls. The control module **706** may be provided in the form of a central controller, a repair control system, or a placement control system. The control module **706** can be accessed through the user interface on the autonomous robotic platform **100**. The control module **706** may be communicatively coupled to the notification module **711**, the sensor module **712**, the communication module **707**, the safety system module **715**, the command module **716**, a disconnect/power switch, or a combination thereof.

[0050] In some aspects, the control module **706** may be configured to route and/or schedule the movement of the autonomous robotic platform **100** to prevent collisions and facilitate efficient repairs of the one or more spalls. In some embodiments, the control module **706** can monitor and control one or more system components **700** of the autonomous robotic platform **100**. In one non-limiting example, the control module **706** can monitor a pressure of the autonomous robotic platform **100** or a system component **700** (or receive pressure data from the sensor module **712**) and send a command to a compressor associated with one or more of the material-holding containers to activate a nozzle when a pressure level drops below a predefined threshold value. In some embodiments, the control module **706** can receive sensor data from the one or more sensors of the sensor arrays to monitor a parameter of the system (e.g., temperature, pressure, battery charge, vibration, current, voltage, etc.). The control module **706** can also monitor a temperature of the system **100**, or a system component **700** and initiate a ventilation system associated with the system **100**, if the temperature value exceeds a predefined temperature threshold value.

[0051] The control module **706** is further designed to communicate either directly or indirectly with a data store **718**, a network **719**, and the mechanical module **710** via one or more wired and/or wireless communication protocols.

[0052] The network **719** includes, for example, the Internet, intranets, extranets, wide area networks (“WANS”), local area networks (“LANS”), wired networks, wireless networks, cloud networks, or other suitable networks, or any combination of two or more such networks. For example, the network **719** can include satellite networks, cable networks, Ethernet networks, and other types of networks. In one embodiment, the network **719** is an isolated private network utilizing a private IP address and limiting access to the network **719**. In some embodiments, the network **719** can include one or more computing devices or storage devices that can be arranged, for example, in one or more server banks or computer banks, or other arrangements. Such devices may host the data store **718** and/or the control module **706**.

[0053] The data store **718** may be provided in the form of a database, a look-up table, a memory unit, or any other suitable data storage medium. The data store **718** may store various types of data including, for example, site-specific instructions for repairing one or more spalls, spall repair specifications based on surface material being repaired, a time-stamped log of the one or more tasks performed by the autonomous robotic platform **100**, and other parameters associated with the one or more autonomous robotic platforms and tasks associated therewith (e.g., serial numbers, firmware versions, GPS locations, battery levels, maintenance intervals, etc.). In addition, the data store **718** may store commands or programmable instructions for the control module **706** to execute. The data store **718** is also designed to receive and update the stored data based on communications received from the control module **706** and other devices/components connected to the data store **718** via the network **719**. In some aspects, the data store **718** is designed to provide

data to the control module **706** or other system components **700** communicatively coupled via the network **719**.

[0054] The safety system module **715** can be configured to integrate one or more dynamic robotic zoning systems, which define a detailed three-dimensional work zone for the autonomous robotic platform **100** to travel throughout. In some embodiments, the safety system module **715** can be configured to detect and/or signal to operators located in a work zone to mitigate accidents, collisions, or obstacles. In some forms, the safety system module **715** can communicate with the vision system module **717** for image processing, hazard detection, and obstacle avoidance. The safety system module **715** may be configured to recognize and generate commands based on relevant national safety standards such as NFPA 70, NFPA 79, UL 508A, OSHA, and relevant articles under regulations and standards 29 CFR 1810.

[0055] In some embodiments, safety system module **715**, through the control module **706**, utilizes a disconnect switch to isolate energy from any electrical systems or cables and safety interlocks of the autonomous robotic platform **100** when the machine is opened. In at least this way, the safety system module **715** de-energizes the electrical cabinet(s) and other electrical components to allow the operator **104** to perform maintenance, repairs, testing, and similar on the platform without exposure to hazardous voltages or currents. The safety system module **715** may also be electronically configured with an e-stop switch, which may prevent damage to the hardware and software of the autonomous robotic platform **100** in the event of a collision, component malfunction, or other system issue.

[0056] In one embodiment, the safety system module **715** can further include a lightning rod and a ground brush, to enhance the electrical safety of the autonomous robotic platform **100** and protect the components of the autonomous robotic platform system **700** in the event of a short circuit, lightning event, ground fault, or similar. Additionally, all electrical components of the system **700** can include circuit protection devices and/or grounding. The circuit protection devices may be provided in the form of fuses, circuit breakers, or other types of overcurrent protection devices. In some embodiments, electronic overcurrent protection devices may be used to supplement the hardware-based overcurrent devices.

[0057] In addition to the safety system module **715**, some embodiments of the autonomous robotic platform **100** may employ a vision system module **717**. The vision system module **717** can be provided in the form of a LiDAR system, camera system, and/or a sensor suite including one or more sensing devices. The vision system module **717** is designed to use one or more computer vision systems and/or machine learning algorithms to identify objects, obstacles, spalls, and perform analysis of the spalls. The vision system module **717** can also be used to accurately identify a spall and one or more parameters associated with the spall (e.g., size, depth, surface material, type of damage, condition of the surface, etc.) to generate or otherwise process a repair specification for the spall(s).

[0058] In some embodiments, the vision system module **717** can identify the unique identifier for each autonomous robotic platform **100** of a group of autonomous robotic platforms (or other autonomous or semi-autonomous machines or devices) and initiate a coordinating group of robots to communicate with one another to effectively and simultaneously repair multiple spalls. For example, when tasked with a multi-repair project, the autonomous robotic platform **100** may communicate and work with other autonomous robotic platforms to efficiently and effectively repair multiple spalls in a specified area. In some embodiments, the vision system module **717** may be configured to detect one or more fiducial tags placed on the operator **104** and/or on other autonomous machines. The fiducial tags can be provided in the form of a QR code, a metallic tag, a plastic tag, a barcode, a number, a digital code, an RFID tag, an LED tag, or other type of unique identifier. In some embodiments, the fiducial tags can be provided in the form of an identifier that can be detected by the system's camera system regardless of the brightness and/or shadow in the environment. In some forms, image processing techniques can be used to detect and identify the

fiducial tag or unique identifier.

[0059] The vision system module **717** is designed to communicate with the navigation module **713** to execute advanced navigation processes of the one or more autonomous machines. The vision system module **717** and/or the navigation module **713** may perform object recognition and other image processing techniques to detect and avoid safety hazards (e.g., obstacles, debris, vehicles, humans, etc.) in a field of view and/or projected travel path of the autonomous robotic platform **100**. In some embodiments, the vision system module **717** can receive and/or process data from an onboard odometer of the autonomous robotic platform **100**. The odometry data can be used by the navigation module **713** in planning a path for the autonomous robotic platform **100**. The navigation module **713** can further include a GPS sensor or similar geo-positioning device. In some aspects, the navigation module **713** can implement geo-fencing or similar techniques to limit a range of motion and/or speed of the autonomous robotic platform **100** within a work zone.

[0060] The command module **716** is designed to receive and/or retrieve inputs associated with the one or more tasks assigned to the autonomous robotic platform **100**. For example, the command module **716** may receive and/or retrieve inputs from operator input, a remote controller, a remote server, or similar. In some aspects, the one or more tasks may refer to repairing one or more spalls. The command module **716** is communicatively coupled to the control module **706**, the sensor module **712**, the navigation module **713**, the electrical module **714**, the vision system module **717**, the power module **708**, the construction module **709**, and the mechanical module **710**. The command module **716** may also be communicatively coupled with the notification module **711** to display one or more notifications associated with a task, like that a task has been assigned or an indication of progress status for a current repair assignment, for example. The notification module **711** may transmit a progress status for a current assignment to the operator, remote controller, remote server, or other user device using the network **719**.

[0061] The system components **700** can further include a power module **708** for one or more of the autonomous robotic platforms. The power module **708** can include the generator **205** and the rechargeable battery (see FIG. 2). The power module **708** can also include a battery management system. The battery management system is designed to monitor the batteries, control battery charging, report a battery status, and intelligently adjust charging configurations to increase battery life. The battery management system can control one or more charging processes by setting a charge rate for the battery charger(s), monitoring and/or modifying one or more battery charger parameters, and disabling a battery charger if a defect is detected. The battery management system can also control the charge and discharge of the battery pack(s) to maximize battery pack longevity and minimize damage to the battery pack(s). The battery management system can also control the main battery discharge contactor if the system detects a hazardous situation or fault condition (e.g., overcurrent, overheat, overvoltage, leak, short circuit, etc.). When the main battery discharge is activated, a disconnect switch is opened and the battery pack is disconnected from the system components **700**. The battery management system can also generate one or more reports related to a status of the battery pack (e.g., charge level, discharge level, fault, etc.). In some embodiments, the power module **708** can further include a power switch for the autonomous robotic platform **100**, which controls power provided to one or more of the system components **700**.

[0062] The construction module **709** includes materials for operation of the autonomous robotic platform **100**. For example, the construction module **709** can include water, repair material, local aggregate, binding agent(s), special solutions, and/or combination thereof. The construction module **709** can further include the hopper **105**, the liquid tank **108**, the binding agent tank **112**, and other containers or similar holding devices for storing and dispensing repair materials for the spall repair processes. The construction module **709** includes the binding agent nozzle **115** and the liquid nozzle **116** for dispensing the binding agent and the liquid from the binding agent tank **112** and the liquid tank **108**, respectively. In some examples, additional nozzles or dispensing devices may be provided.

[0063] The mechanical module **710** includes the track component(s) **113**, the plow attachment **114**, the auger **117**, the plate compactor attachment **307**, the motor **306** and other aspects of the autonomous robotic platform **100** described in connection with FIGS. **1-6**. In some embodiments, the mechanical module **710** is communicatively coupled to the control module **706** to execute operations using aspects of the mechanical module **710** to complete the spill repair assignment.

[0064] FIGS. **8A-8F** illustrate non-limiting example of spill repair process **800** executed by the autonomous robotic platform **100** of FIG. **1**. The process **800** includes several repair steps being performed over a relatively short time interval (e.g., 45 minutes). The process steps include identifying and pushing debris into the spill (FIG. **8A**), flooding the spill (FIG. **8B**), applying repair material to the spill (FIG. **8C**), applying bulk material to the spill (FIG. **8D**), applying repair material a second time to the spill (FIG. **8E**), and compacting the repair material (FIG. **8F**).

[0065] As shown in FIG. **8A**, the autonomous robotic platform **100** may begin the spill repair process **800** by executing programmable instructions to retrieve or otherwise process input coordinates and navigate to the spill **808**. For example, the autonomous robotic platform **100** may retrieve the input coordinates from the operator **104** via a user interface, the data store **718**, the notification module **711**, the communication module **707**, and/or the network **719** (see FIG. **7**).

[0066] In some forms, the navigation module **713** of the autonomous robotic platform **100** generates a route to the spill **808**. The navigation module **713** may receive signals of GPS coordinates and data or information from the vision system module **717** and the notification module **711** to travel to the spill area. In some embodiments, the autonomous robotic platform **100** receives manually input coordinates and uses automatic object detection via the vision system module **717** to navigate to the spill location. In some embodiments, the notification module **711** sends a release signal to a user device or other computing device to indicate that the autonomous robotic platform has reached the location coordinates for the spill repair assignment.

[0067] In some embodiments, the autonomous robotic platform **100** measures a size and/or volume of the spill **808** and determines the amount and type of repair material **806** that should be used to complete the repair using the sensor module **712**, the vision system module **717**, or a combination thereof. In some embodiments, the vision system module **717** can also be used to accurately identify a spill and one or more parameters associated with the spill (e.g., size, depth, surface material, type of damage, condition of the surface, etc.) to generate or otherwise process a repair specification for the spill(s). The autonomous robotic platform **100** may identify that debris exists around an exterior of the spill **808** and push the debris **807** in the spill **808** as the first step in the repair process. Using existing debris surrounding the spill **808** can not only prepare the surface area around the spill **808** to be sealed smoothly but can also reduce the amount of repair material used to complete the repair since a portion of the spill volume will be filled using the existing debris. In some embodiments, the vision system module **717** can re-assess the volume of the spill after the debris is pushed into the spill. In some aspects, the debris **807** is moved using the plow attachment **114**. At the process step illustrated in FIG. **8B**, the autonomous robotic platform **100** floods the spill **808** with a liquid **805** (e.g., water) from the liquid tank **108** via the liquid nozzle **116**. It will be appreciated that some repair materials may not utilize a liquid as part of the repair process, according to the specification for the repair and the recommended usage for the repair material.

[0068] The process step illustrated in FIG. **8C** includes applying the repair material **806** into the spill **808**. In some embodiments, the repair material **806** is applied using a nozzle. In some aspects, the liquid (e.g., water) is flooded into the spill **808** and a coat of binding agent is applied before the spill **808** is filled with repair material **806**. As shown in FIG. **8D**, the autonomous robotic platform **100** applies a bulky material **809** to the spill **808** using the plow attachment **114**.

[0069] In the process step shown in FIG. **8E** the autonomous robotic platform **100** re-applies the repair material **806** to the spill **808** and determines, through the vision system module **717**, if the spill **808** is appropriately filled or if additional repair materials should be dispensed. When the

spall **808**, is filled, as shown in FIG. 8F, the autonomous robotic platform **100** finalizes the spall repair using the plate compactor attachment **307** (e.g., tamper) and generates a notification to be sent via the notification module **711** that the repair assignment is complete. In some embodiments, the vision system module **717** may verify the spall **808** is repaired according to a repair specification (either identified automatically using one or more trained models of the processing module **705**, or by manual input, or as otherwise determined by the platform **100**) before sending the notification. The notification, in some aspects, includes an image, location, and/or description of the spall **808** repaired. As discussed above, the example process steps shown in FIGS. 8A-8F are non-limiting and some spall repairs may not utilize one or more of the construction materials indicated.

[0070] FIG. 9 is a block diagram of a method **900** for repairing spalls using the autonomous robotic platform of FIG. 1. The method **900** utilizes the system components **700** from FIG. 7. To begin with, the method **900**, may start at **905** to process coordinates associated with a spall assigned for repair and navigate to the spall. In some embodiments, the navigation module **713**, the control module **706**, and the vision system module **717** (see FIG. 7) may process the coordinates and execute one or more aspects of the navigation step **905**.

[0071] At step **906**, the autonomous robotic platform **100** prepares the spall for repair. In some aspects, preparing the spall for repair can include removing debris from an area around the spall, as shown in FIG. 8A. In some embodiments, the step for preparing the spall for repair may include removing debris from the spall, depending on the repair specification for the ground surface, type of spall, and repair material installation instructions.

[0072] At step **907**, the autonomous robotic platform **100** measures the size and/or volume of the spall and determines an amount of repair material to be used for the repair. The volume analysis may be performed by the vision system module **717**, the sensor module **712**, the control module **706**, or a combination thereof.

[0073] At step **908**, the autonomous robotic platform **100** disperses the appropriate amount of repair material into the spall and applies one or more layers of binding agent to the newly added repair material (if binding agent is specified based on the specific repair material being used).

[0074] At step **909**, the autonomous robotic platform **100** finalizes the spall repair by tampering the repair material to create a smooth surface using the plow attachment **114**. The notification module **711** can generate and transmit a notification of the status/completion of the spall repair assignment. The notification can include metadata (e.g., data related to the type of spall, damage information, spall location, time stamps for repair start and completion times, amount of repair material, time spent performing each step in the repair process, weather or other environmental conditions during the repair, etc.), image information, and a detailed completion report associated with the repair of the spall.

[0075] The autonomous robotic platform **100** and the system components **700** may be adapted to generate, train, and execute a plurality of trained learning models, nodes, neural networks, gradient boosting algorithms, mutual information classifiers, random forest classifications, and other machine learning- and artificial intelligence-related algorithms to process the parameters, features, and other data elements. In some embodiments, the one or more trained learning models can include deep learning, machine learning, neural networks, vision, and similar advanced artificial intelligence-based technologies. When used throughout the present disclosure, one skilled in the art will understand that processes for iteratively training the “trained learning model” can include machine learning processes and other advanced artificial intelligence processes. For example, the system components **700** and processes of the present disclosure can perform data processing, image analysis, generate tasks or action items, provide customized recommendations according to user settings and preferences, generate interfaces, generate reports, generate files, generate notifications, and similar processes. In some embodiments, the system components **700** may use additional inputs and/or feedback loops to an iterative training process for a personalized event hosting

process based on a plurality of parameters and adjustable metric values.

[0076] Where used herein, the user device and/or computing device can be any device capable of connecting to the Internet. In one embodiment, the user device is provided in the form of a portable device like a smartphone, tablet, laptop, other computing device, or other display interface. In some embodiments, the user device can include, or is otherwise connected to, a programmable processor, a network interface, one or more data capture devices, and a memory unit. In some embodiments, the data capture devices can be provided in the form of one or more of a microphone, a camera, a vibration sensor, an accelerometer, and/or other sensing or recording device. In some embodiments, the user device can include a Wi-Fi, Bluetooth, cellular, wired connection, wireless connection, or similar communication link used to communicate with the network **719**. In some embodiments, the user device can include programmable instructions that are stored on a user device non-transitory computer readable medium and/or in the data store **718** and executed by the programmable processor to perform one or more of the processes described herein.

[0077] The previous description of the disclosed embodiments is provided to enable any person skilled in the art to make, use, or practice the disclosure. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the disclosure. Thus, the disclosure is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

[0078] Specific embodiments of the system and method for repairing a spall according to the present disclosure have been described for the purpose of illustrating the manner in which the disclosure can be made and used. It should be understood that the implementation of other variations and modifications of this disclosure and its different aspects will be apparent to one skilled in the art, and that this disclosure is not limited by the specific embodiments described. Features described in one embodiment can be implemented in other embodiments. The subject disclosure is understood to encompass the present invention and any and all modifications, variations, or equivalents that fall within the spirit and scope of the basic underlying principles disclosed and claimed herein.

Claims

1. An autonomous robotic platform for repairing a spall comprising: a power module; a construction module including a tank for holding a repair material; a mechanical module including a track component for moving the autonomous robotic platform and a plow attachment for moving debris proximate to the spall to be repaired; a processing module comprising: a vision system module designed to identify and analyze the spall; a control module to execute one or more programmable instructions of the autonomous robotic platform; and a navigation module.
2. The autonomous robotic platform of claim 1, wherein the processing module further comprises: a notification module; a sensor module; a communication module; and an electrical safety system.
3. The autonomous robotic platform of claim 2, wherein the vision system module further comprises: a camera system; a sensor system; and one or more computer vision algorithms.
4. The autonomous robotic platform of claim 1, wherein the navigation module is configured to process location coordinates associated with a spall repair assignment and navigate the autonomous robotic platform to the location coordinates.
5. The autonomous robotic platform of claim 2, wherein the control module further comprises a disconnect switch.
6. The autonomous robotic platform of claim 1, wherein the power module includes a generator, a battery, a battery management system, or a combination thereof.
7. The autonomous robotic platform of claim 4, wherein a notification module sends a release

signal to indicate that the autonomous robotic platform has reached the location coordinates.

8. The autonomous robotic platform of claim 2, wherein the electrical safety system is configured to limit operations of the autonomous robotic platform in a defined three-dimensional work zone for a spall repair assignment.

9. An autonomous robotic platform designed for repairing one or more spalls comprising: a power module; a construction module including a tank for holding a repair material and a nozzle for dispensing the repair material; a mechanical module including a track component for moving the autonomous robotic platform; a processing module comprising: a vision system module designed to identify and analyze the spall; and a navigation module designed to route the autonomous robotic platform to a location coordinate associated with the spall.

10. The autonomous robotic platform of claim 9, wherein the power module includes a generator, a rechargeable battery, a battery management system, or a combination thereof.

11. The autonomous robotic platform of claim 9, wherein the mechanical module further comprises: a plow attachment for moving debris proximate to the spall to be repaired; and a plate compactor attachment to tamper the spall after the repair material is applied to the spall.

12. The autonomous robotic platform of claim 9, wherein the tank for holding a repair material is provided in the form of a liquid tank, a binding agent tank, a hopper, or a combination thereof.

13. The autonomous robotic platform of claim 9, wherein the autonomous robotic platform further comprises a user interface to send and retrieve information from a data store and a network communicatively coupled to the autonomous robotic platform.

14. A method of repairing a spall using an autonomous robotic platform, the method comprising: providing location coordinates associated with a spall repair assignment to a navigation module of the autonomous robotic platform; navigating to the location coordinates using a vision system module; preparing the spall for a repair using a plow attachment to move debris proximate to the spall; analyzing a volume of the spall using the vision system module; determining an amount of repair material to be used for the spall; dispersing the amount of repair material determined into the spall; finalizing the repair assignment; and generating a notification to a user device that the spall repair assignment is complete.

15. The method of claim 14, wherein the location coordinates are provided in the form of GPS coordinates.

16. The method of claim 14, further comprising: applying a liquid to the spall from a liquid tank of the autonomous robotic platform.

17. The method of claim 14, wherein preparing the spall for the repair further comprises moving the debris into the spall and analyzing the volume of the spall after the debris has been moved into the spall.

18. The method of claim 14, wherein preparing the spall for the repair includes flooding the spall with liquid.

19. The method of claim 18, wherein the liquid is water.

20. The method of claim 14, wherein the notification includes metadata and an image of the spall after the repair.
